

PPITRACKER-61

PipelineParameter Modification - Implementation and Backend Handoff

Current result: The PPI Breakdown desktop implementation is complete. Testing proves that the tracker backend returns HTTP 200 but removes `config.parameter_index`, preventing persistent storage.

1. Requested Behavior

- Add `pipeline_parameter` to Add Column and Edit Column data-type options.
- Show a Pipeline Parameter selection combo box only for that data type.
- Store the selected PipelineParameter ID as `config.parameter_index`.
- Load values for each asset and shot location from SQLite by default.
- Allow a checkable File menu option to switch retrieval to CentralWeb.
- Display PipelineParameter columns as read-only.

2. Work Completed in PPI Breakdown

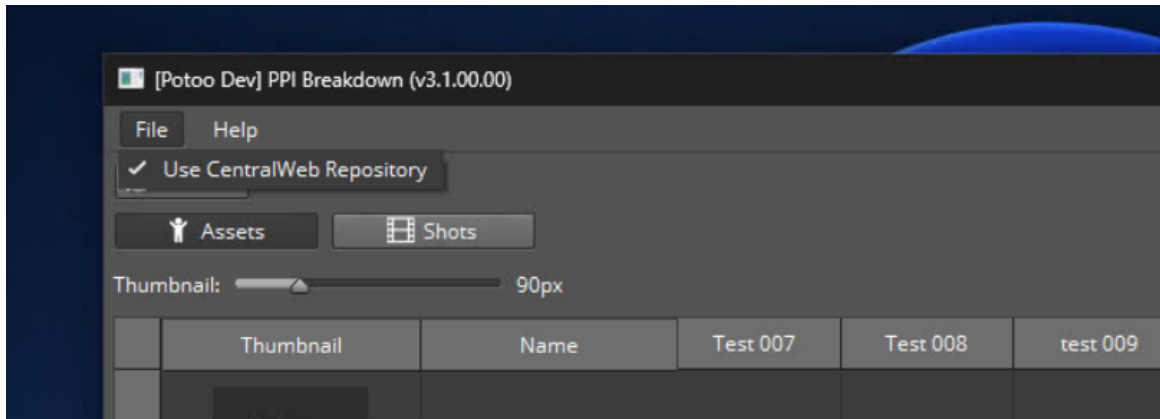
Area	Status	Details
Column UI	Complete	Added the data type, conditional selection combo, parameter loading, edit restoration, and validation.
API client	Complete	Create and update payloads include <code>config.parameter_index</code> for <code>pipeline_parameter</code> columns.
SQLite	Complete	SQLiteRepository is the default source for parameter definitions and values.
CentralWeb	Complete	Added a checkable File menu action and repository switching with cache invalidation.
Table display	Complete	Asset and shot locations resolve to PipelineParameter values; output is read-only.
CSV and editing	Complete	Values can be exported and are protected from direct editing/import updates.

Files Updated

- `src/ppiTracker/columnManagerDialog.py` - selection UI, validation, save/edit restoration.
- `src/ppiTracker/apiClient.py` - create/update payload support.
- `src/ppiTracker/service.py` - SQLite/CentralWeb repositories, caches, and repository switch.
- `src/ppiTracker/mainWindow.py` - menu toggle, display lookup, refresh, and read-only behavior.

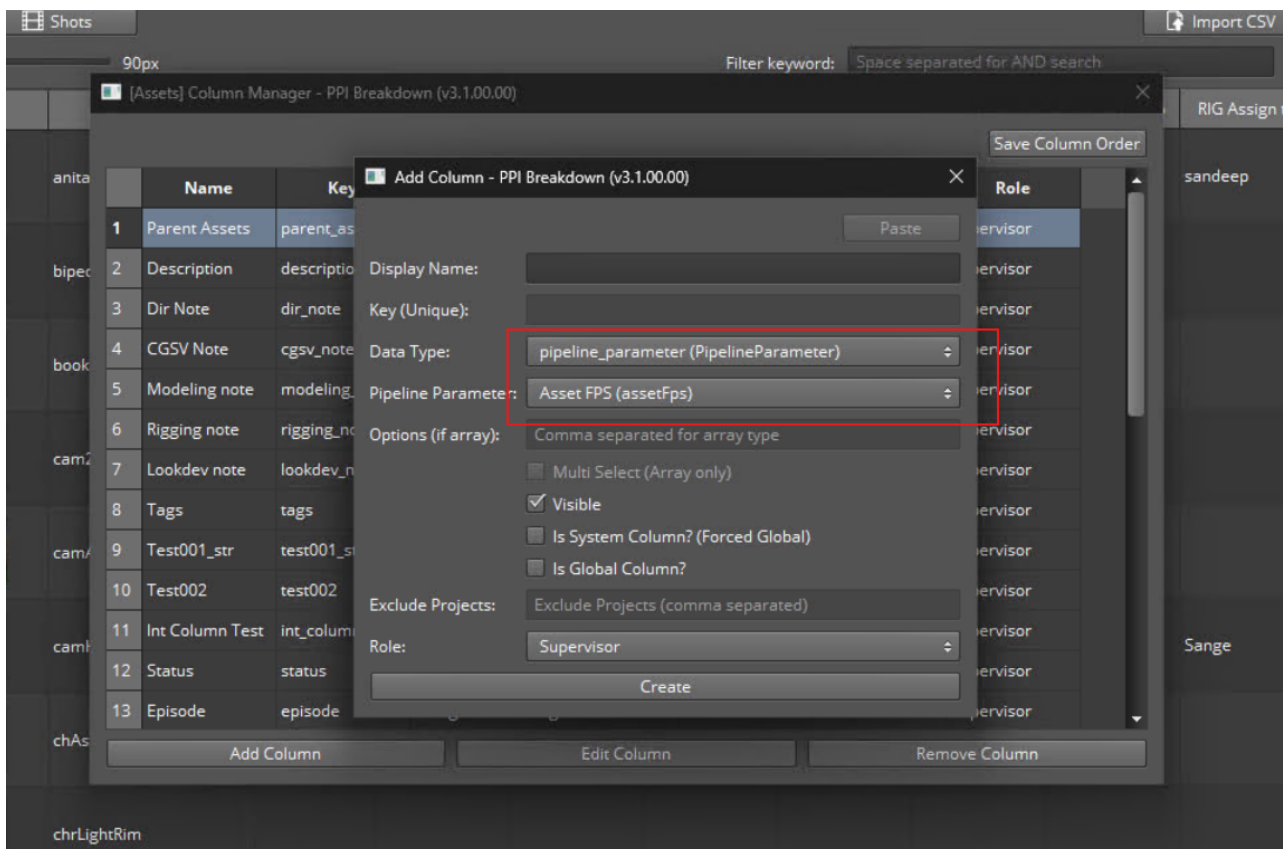
3. Implemented Options - Screenshots

Use CentralWeb Repository



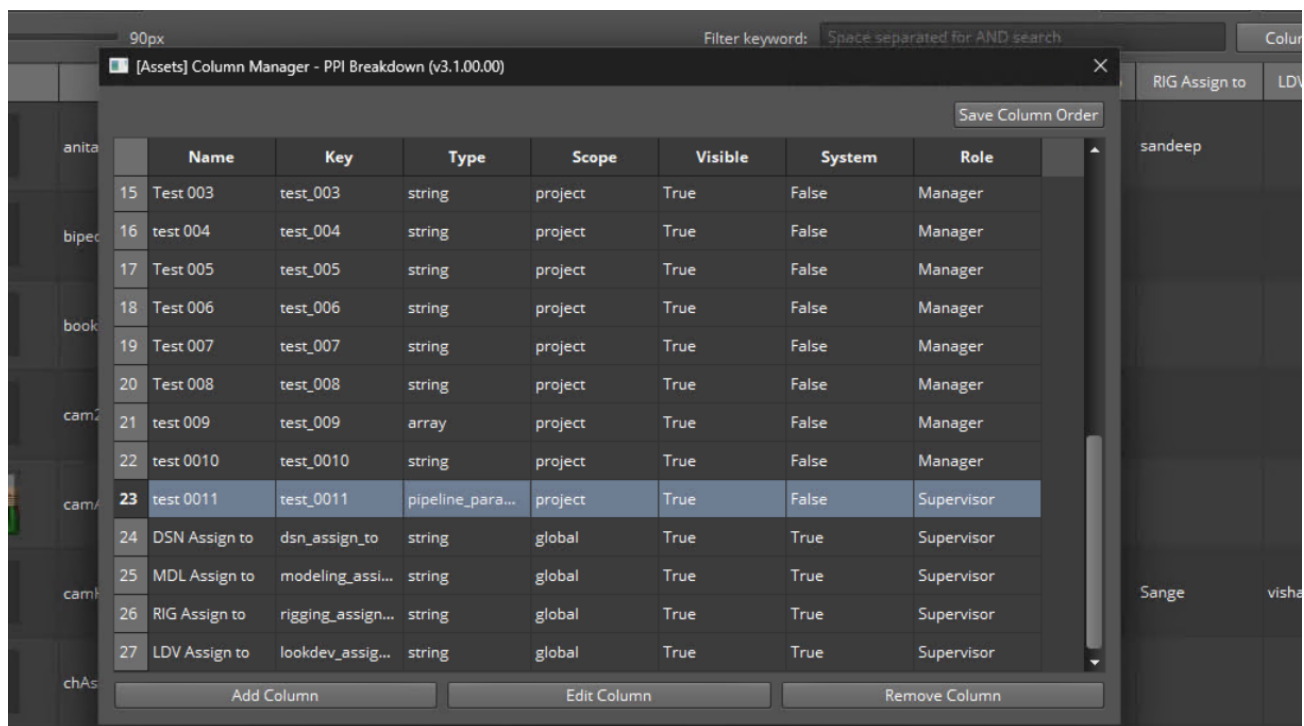
This checkable File-menu option selects the PipelineParameter value source. Unchecked uses the studio SQLite database, which is the default. Checked uses the CentralWeb repository. Changing the option clears cached parameter values and refreshes the table.

PipelineParameter column setup



Selecting pipeline_parameter as the Data Type displays the Pipeline Parameter combo box. The user chooses a parameter such as Asset FPS. The selected PipelineParameter ID must be stored as config.parameter_index.

Saved PipelineParameter column



90px Filter keyword: Space separated for AND search

[Assets] Column Manager - PPI Breakdown (v3.1.00.00)

Save Column Order

	Name	Key	Type	Scope	Visible	System	Role
15	Test 003	test_003	string	project	True	False	Manager
16	test 004	test_004	string	project	True	False	Manager
17	Test 005	test_005	string	project	True	False	Manager
18	Test 006	test_006	string	project	True	False	Manager
19	Test 007	test_007	string	project	True	False	Manager
20	Test 008	test_008	string	project	True	False	Manager
21	test 009	test_009	array	project	True	False	Manager
22	test 0010	test_0010	string	project	True	False	Manager
23	test 0011	test_0011	pipeline_para...	project	True	False	Supervisor
24	DSN Assign to	dsn_assign_to	string	global	True	True	Supervisor
25	MDL Assign to	modeling_assi...	string	global	True	True	Supervisor
26	RIG Assign to	rigging_assign...	string	global	True	True	Supervisor
27	LDV Assign to	lookdev_assig...	string	global	True	True	Supervisor

Add Column Edit Column Remove Column

After creation, the column appears in Column Manager with type pipeline_parameter. Its values are displayed for matching Asset or Shot locations and are read-only in the main table.

4. Verified Failure Evidence

When column test_0011 was edited and saved, the desktop application sent PipelineParameter ID 193. The API returned status 200, but the response omitted it.

```
[INFO][2026-09-25 14:20:33.649][16208]ppiTracker.mainWindow: Assets rows loaded at 4.61s (C:\SanjayK\tracker30\devBranch\tracker30\src\ppiTracker\mainWindow.py:_logElapsed:766)
[INFO][2026-09-25 14:20:34.165][16208]ppiTracker.mainWindow: Assets thumbnails dispatched at 5.13s (C:\SanjayK\tracker30\devBranch\tracker30\src\ppiTracker\mainWindow.py:_logElapsed:766)
UPDATE COLUMN PAYLOAD: {'role': 'supervisor', 'role_setting': 'supervisor', 'project': 'potoodev', 'display_name': 'test_0011', 'data_type': 'pipeline_parameter', 'config': {'options': [], 'multi_select': False, 'role': 'supervisor', 'role_setting': 'supervisor', 'parameter_index': 193, 'visible': True, 'scope': 'project', 'exclude_projects': [], 'is_system': False, 'user': 'sanjay', 'root': 'assets', 'key': 'test_0011'}}
UPDATE COLUMN RESPONSE: 200 {'display_name': 'test_0011', 'data_type': 'pipeline_parameter', 'config': {'options': [], 'multi_select': False, 'visible': True, 'scope': 'project', 'exclude_projects': [], 'is_system': False, 'user': 'sanjay', 'studio': ''}}
Getting pipeline parameter column value for column: test_0011 {'options': [], 'multi_select': False, 'role': 'supervisor', 'role_setting': 'supervisor'} and group: anitalHero
Getting pipeline parameter column value for column: test_0011 {'options': [], 'multi_select': False, 'role': 'supervisor', 'role_setting': 'supervisor'} and group: bipedHero
Getting pipeline parameter column value for column: test_0011 {'options': [], 'multi_select': False, 'role': 'supervisor', 'role_setting': 'supervisor'} and group: bipedHero
```

Runtime evidence: the update payload contains parameter_index 193. The HTTP 200 response returns the pipeline_parameter column but omits parameter_index from config.

Request sent by PPI Breakdown

```
{
  'data_type': 'pipeline_parameter',
  'config': {
    'options': [],
    'multi_select': False,
    'role': 'supervisor',
    'role_setting': 'supervisor',
    'parameter_index': 193
  }
}
```

Successful API response

```
HTTP 200

{'data_type': 'pipeline_parameter',
 'config': {
   'options': [],
   'multi_select': False
 }}

# parameter_index is missing
```

Conclusion: PPI Breakdown sends the correct field. The API's request model, serialization model, or persistence layer filters it out before storing/returning the column.

5. Required Backend Fix

- Update tracker-column create and update request schemas to accept config.parameter_index as an optional integer.
- Preserve the field when validating and serializing the config object.
- Persist the field in the column's database config JSON.
- Return the field from create, update, and column-list responses.
- Keep the field optional for all existing non-PipelineParameter columns.

Affected endpoints

```

POST /api/projects/{project}/tracker/columns
PUT  /api/projects/{project}/tracker/columns/{columnID}
GET  /api/projects/{project}/tracker/columns

```

Expected stored/returned structure

```

{
  "data_type": "pipeline_parameter",
  "config": {
    "parameter_index": 193
  }
}

```

6. Backend Acceptance Criteria

- Creating a pipeline_parameter column with parameter_index returns the same ID.
- Updating the selected parameter replaces the stored ID.
- Reloading the column list still returns parameter_index.
- Existing string, array, date, bool, and float columns remain unchanged.
- An absent parameter_index remains valid for non-pipeline data types.
- Invalid non-integer IDs are rejected with a clear 4xx validation response.

7. Retest Steps After Backend Deployment

	Start PPI Breakdown with project potoodev.
	Open Column Manager and edit or create a pipeline_parameter column.
	Select a Pipeline Parameter and save the column.
	Close and reopen the dialog; confirm the selection is restored.
	Refresh the main table; confirm values display for matching assets/shots.
	Enable File > Use CentralWeb Repository and refresh.
	Confirm CentralWeb values display and the column remains read-only.
	Restart the application and confirm the configuration still works.

8. Recommended Ticket Note

PPI Breakdown sends config.parameter_index correctly for pipeline_parameter columns. The tracker column API returns HTTP 200 but strips the field from the response and stored config. Please update the create/update/list schemas and persistence logic to accept and retain parameter_index as an optional integer. Reproduction: request ID 193 is present; response config contains only options and multi_select.